2020 AMC 10A Problem 9 - Solution

Review the full statement and step-by-step solution for 2020 AMC 10A Problem 9. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 9

Problem:

A single bench section at a school event can hold either 7 adults or 11 children. When N bench sections are connected end to end, an equal number of adults and children seated together will occupy all the bench space. What is the least possible positive integer value of N?

Answer Choices:

A. 9
B. 18
C. 27
D. 36
E. 77

Solution:

The least common multiple of 7 and 11 is 77 . Therefore, there must be 77 adults and 77 children.

The total number of benches is

$$\frac{77}{7}+\frac{77}{11}=11+7=\boxed{(B)\ 18}.$$

$$\textbf{OR}$$

Let x denote how many adults there are.

Since the number of adults is equal to the number of children we can write N as

$$\frac{x}{7}+\frac{x}{11}=N$$

Simplifying we get

$$\frac{18x}{77}=N$$

Since both n and x have to be positive integers, x has to equal 77. Therefore, $N=\boxed{(B)\ 18}$ is our final answer.